New Wheels Project

Introduction to SQL

Problem Statement

Business Context

A lot of people in the world share a common desire: to own a vehicle. A car or an automobile is seen as an object that gives the freedom of mobility. Many now prefer pre-owned vehicles because they come at an affordable cost, but at the same time, they are also concerned about whether the after-sales service provided by the resale vendors is as good as the care you may get from the actual manufacturers.

New-Wheels, a vehicle resale company, has launched an app with an end-to-end service from listing the vehicle on the platform to shipping it to the customer's location. This app also captures the overall after-sales feedback given by the customer.

Objective

New-Wheels sales have been dipping steadily in the past year, and due to the critical customer feedback and ratings online, there has been a drop in new customers every quarter, which is concerning to the business. The CEO of the company now wants a quarterly report with all the key metrics sent to him so he can assess the health of the business and make the necessary decisions.

As a data analyst, you see that there is an array of questions that are being asked at the leadership level that need to be answered using data. Import the dump file that contains various tables that are present in the database. Use the data to answer the questions posed and create a quarterly business report for the CEO.

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Business Questions

Question 1: Find the total number of customers who have placed orders. What is the distribution of the customers across states?

Solution Query:

```
SELECT
   STATE,
   COUNT(DISTINCT CUSTOMER_ID) AS CNT
FROM
   customer_t
WHERE
   CUSTOMER_ID IN (
      SELECT CUSTOMER_ID
      FROM order_t
   )
GROUP BY
   STATE
ORDER BY
    CNT DESC;
```

Output:

```
        SELECT
        STATE,
        COUNT(DISTINCT CUSTOMER_ID) AS CNT
    FROM
        customer_t
    WHERE
        CUSTOMER_ID IN (
            SELECT CUSTOMER_ID
            FROM order_t
        )
    GROUP BY
        STATE
    ORDER BY
         CNT DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

STATE	CNT
California	97
Texas	97
Florida	86
New York	69
District of Columbia	35
Colorado	33
Ohio	33
Alabama	29

Result 1

Output

Action Output

#	Time	Action	Message
1	22:26:10	use vehdb	0 row(s) affected
2	22:26:10	SELECT STATE, COUNT(DISTINCT CUSTOMER_ID) AS CNT FROM customer_t WHE...	49 row(s) returned

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Observations and Insights:

- California and Texas has the highest number of customers(97).
- Followed by Florida(86) and New York(69).

Question 2: Which are the top 5 vehicle makers preferred by the customers?

Solution Query:

```
SELECT
   P.VEHICLE_MAKER,
   COUNT(O.CUSTOMER_ID) AS CNT
FROM
   product_t P
JOIN
   order_t O
   ON P.PRODUCT_ID = O.PRODUCT_ID
GROUP BY
   P.VEHICLE_MAKER
ORDER BY
   CNT DESC
LIMIT 5;
```

Output:

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```sql
SELECT
    P.VEHICLE_MAKER,
    COUNT(O.CUSTOMER_ID) AS CNT
FROM
    product_t P
JOIN
    order_t O
    ON P.PRODUCT_ID = O.PRODUCT_ID
GROUP BY
    P.VEHICLE_MAKER
ORDER BY
    CNT DESC
LIMIT 5;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

VEHICLE_MAKER	CNT
Chevrolet	83
Ford	63
Toyota	52
Pontiac	50
Dodge	50

Result 1

Output

Action Output

#	Time	Action	Message
1	22:00:00	use vehdb	0 row(s) affected
2	22:00:54	SELECT P.VEHICLE_MAKER, COUNT(O.CUSTOMER_ID) AS CNT FROM product_t P JOI...	5 row(s) returned

Observations and Insights:

- Top 5 vehicle markers are Chevrolet, Ford, Toyota, Pontiac and Dodge
- Most preferred one is Chevrolet, followed by Ford.

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Question 3: Which is the most preferred vehicle maker in each state?

Solution Query:

```
SELECT *
FROM (
   SELECT
      STATE,
      VEHICLE_MAKER,
      COUNT(CUSTOMER_ID) AS TOTAL_CUSTOMER,
      RANK() OVER (
         PARTITION BY STATE
         ORDER BY COUNT(CUSTOMER_ID) DESC
      ) AS RANKING
   FROM
      customer_t
      JOIN order_t USING (CUSTOMER_ID)
      JOIN product_t USING (PRODUCT_ID)
   GROUP BY
      STATE,
      VEHICLE_MAKER
) AS V
WHERE
   RANKING = 1
ORDER BY
   TOTAL_CUSTOMER DESC;
```

Output:

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```sql
SELECT *
FROM (
    SELECT
        STATE,
        VEHICLE_MAKER,
        COUNT(CUSTOMER_ID) AS TOTAL_CUSTOMER,
        RANK() OVER (
            PARTITION BY STATE
            ORDER BY COUNT(CUSTOMER_ID) DESC
        ) AS RANKING
    FROM
        customer_t
        JOIN order_t USING (CUSTOMER_ID)
        JOIN product_t USING (PRODUCT_ID)
    GROUP BY
        STATE,
        VEHICLE_MAKER
) AS V
WHERE
    RANKING = 1
ORDER BY
    TOTAL_CUSTOMER DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

STATE	VEHICLE_MAKER	TOTAL_CUSTOMER	RANKING
Texas	Chevrolet	9	1
Florida	Toyota	7	1
California	Ford	6	1
California	Dodge	6	1
California	Audi	6	1
California	Nissan	6	1
California	Chevrolet	6	1
Ohio	Chevrolet	6	1
Alabama	Dodge	5	1

Result 1

Output

Action Output

#	Time	Action	Message
1	22:32:36	use vehdb	0 row(s) affected
2	22:32:36	SELECT * FROM (SELECT STATE, VEHICLE_MAKER, COUNT(CUSTOMER_ID) AS TOTAL_CUSTOMER, ...	143 row(s) returned

Observations and Insights:

- Chevrolet is most preferred one as it's the top selling brand in my states.
- In Texas, it's having its highest customers, with no other brands close to it, followed by California.
- Although market in California is evenly distributed as other companies are also have similar number of customers.

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Question 4: Find the overall average rating given by the customers. What is the average rating in each quarter?

Consider the following mapping for ratings: "Very Bad": 1, "Bad": 2, "Okay": 3, "Good": 4, "Very Good": 5

Solution Query:

```
SELECT
   quarter number,
   ROUND(AVG(Total_Rating), 2) AS Average Rating
FROM (
   SELECT
      customer feedback,
      quarter number,
      CASE
         WHEN customer_feedback = 'Very Bad' THEN 1
         WHEN customer_feedback = 'Bad' THEN 2
         WHEN customer_feedback = 'Okay' THEN 3
         WHEN customer_feedback = 'Good' THEN 4
         WHEN customer_feedback = 'Very Good' THEN 5
      END AS Total_Rating
   FROM
      order_t
) AS R
GROUP BY
   quarter_number
ORDER BY
   quarter_number;
```

Output:

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```sql
SELECT
    quarter_number,
    ROUND(AVG(Total_Rating), 2) AS Average_Rating
FROM (
    SELECT
        customer_feedback,
        quarter_number,
        CASE
            WHEN customer_feedback = 'Very Bad' THEN 1
            WHEN customer_feedback = 'Bad' THEN 2
            WHEN customer_feedback = 'Okay' THEN 3
            WHEN customer_feedback = 'Good' THEN 4
            WHEN customer_feedback = 'Very Good' THEN 5
        END AS Total_Rating
    FROM
        order_t
) AS R
GROUP BY
    quarter_number
ORDER BY
    quarter_number;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

quarter_number	Average_Rating
1	3.55
2	3.35
3	2.96
4	2.40

Result 2

Output

Action Output

#	Time	Action	Message
1	21:42:53	use vehdb	0 row(s) affected
2	21:43:31	SELECT quarter_number, ROUND(AVG(Total_Rating), 2) AS Average_Rating FROM (SELECT customer_feedback, quarter_number, ...	4 row(s) returned
3	21:45:16	SELECT quarter_number, ROUND(AVG(Total_Rating), 2) AS Average_Rating FROM (SELECT customer_feedback, quarter_number, ...	4 row(s) returned

Observations and Insights:

- Customer rating is showing negative trend as it's keep dropping with each passing quarter.
- Q4 has the lowest average rating of 2.4, which is between Okay and Bad.

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Question 5: Find the percentage distribution of feedback from the customers. Are customers getting more dissatisfied over time?

Solution Query:

```
WITH CTE AS (
   SELECT
      quarter_number,
      SUM(CASE WHEN customer_feedback = 'Very Bad' THEN 1 ELSE 0 END) AS Very_Bad,
      SUM(CASE WHEN customer_feedback = 'Bad' THEN 1 ELSE 0 END) AS Bad,
      SUM(CASE WHEN customer_feedback = 'Okay' THEN 1 ELSE 0 END) AS Okay,
      SUM(CASE WHEN customer_feedback = 'Good' THEN 1 ELSE 0 END) AS Good,
      SUM(CASE WHEN customer_feedback = 'Very Good' THEN 1 ELSE 0 END) AS Very_Good,
      COUNT(customer_feedback) AS Feedback
   FROM order_t
   GROUP BY quarter_number
)

SELECT
   quarter_number,
   ROUND((Very_Bad / Feedback) * 100, 2) AS Very_Bad,
   ROUND((Bad / Feedback) * 100, 2) AS Bad,
   ROUND((Okay / Feedback) * 100, 2) AS Okay,
   ROUND((Good / Feedback) * 100, 2) AS Good,
   ROUND((Very_Good / Feedback) * 100, 2) AS Very_Good
FROM CTE
ORDER BY quarter_number;
```

Output:

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```sql
WITH CTE AS (
    SELECT
        quarter_number,
        SUM(CASE WHEN customer_feedback = 'Very Bad' THEN 1 ELSE 0 END) AS Very_Bad,
        SUM(CASE WHEN customer_feedback = 'Bad' THEN 1 ELSE 0 END) AS Bad,
        SUM(CASE WHEN customer_feedback = 'Okay' THEN 1 ELSE 0 END) AS Okay,
        SUM(CASE WHEN customer_feedback = 'Good' THEN 1 ELSE 0 END) AS Good,
        SUM(CASE WHEN customer_feedback = 'Very Good' THEN 1 ELSE 0 END) AS Very_Good,
        COUNT(customer_feedback) AS Feedback
    FROM order_t
    GROUP BY quarter_number
)

SELECT
    quarter_number,
    ROUND((Very_Bad / Feedback) * 100, 2) AS Very_Bad,
    ROUND((Bad / Feedback) * 100, 2) AS Bad,
    ROUND((Okay / Feedback) * 100, 2) AS Okay,
    ROUND((Good / Feedback) * 100, 2) AS Good,
    ROUND((Very_Good / Feedback) * 100, 2) AS Very_Good
FROM CTE
ORDER BY quarter_number;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

quarter_number	Very_Bad	Bad	Okay	Good	Very_Good
1	10.97	11.29	19.03	28.71	30.00
2	14.89	14.12	20.23	22.14	28.63
3	17.90	22.71	21.83	20.96	16.59
4	30.65	29.15	20.10	10.05	10.05

Result 1

Output

Action Output

#	Time	Action	Message
1	21:46:28	use vehdb	0 row(s) affected
2	21:47:02	WITH CTE AS (SELECT quarter_number, SUM(CASE WHEN customer_feedback = 'Very Bad' THEN 1 ELSE 0 END) AS Very_Bad, SUM(CAS...	4 row(s) returned

Observations and Insights:

- It seems like customers are getting dissatisfied over time as there's a declining trend in positive feedback with each passing quarter.
- From the data we can see that from Q1-Q4, there's around 20% jump is "Very bad" feedback and at the same time "Very good" feedback dropped by 20%.

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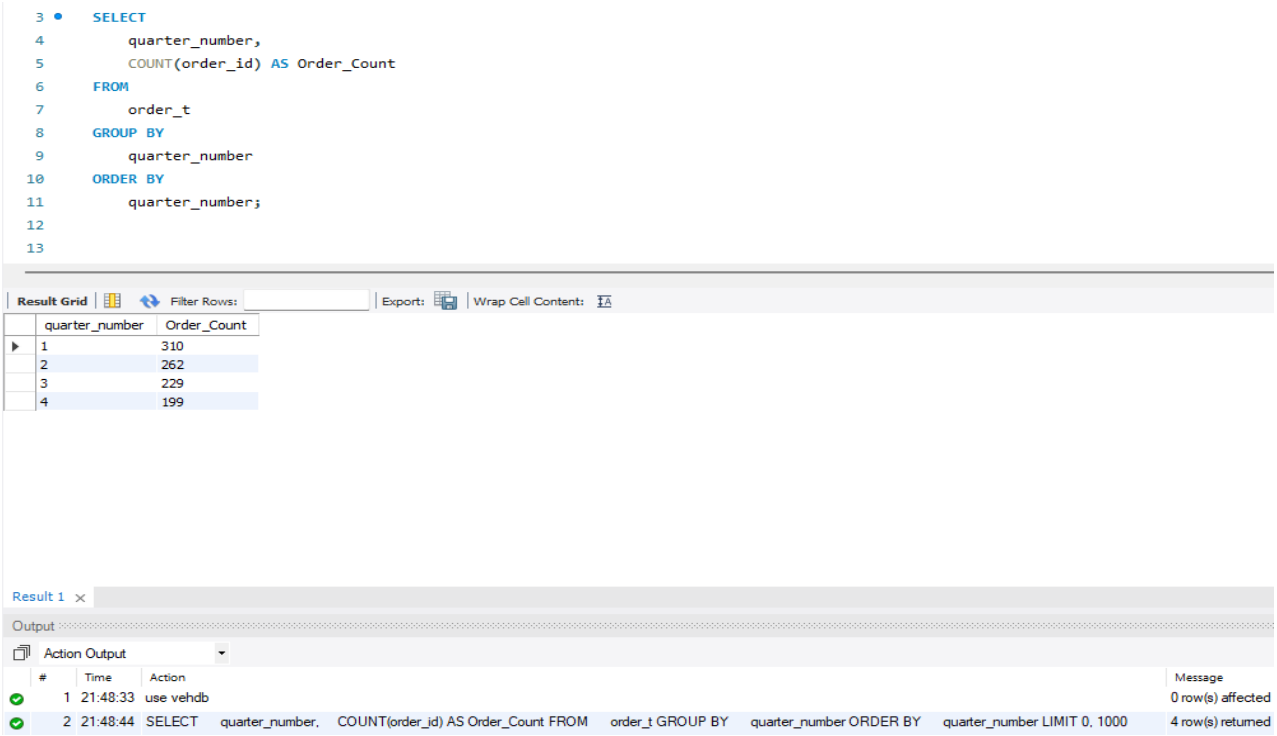
Question 6: What is the trend of the number of orders by quarter?

Solution Query:

```
SELECT
   quarter_number,
   COUNT(order_id) AS Order_Count
FROM
   order_t
GROUP BY
   quarter_number
ORDER BY
   quarter_number;
```

Output:



Observations and Insights:

- There's continuous decline in number of orders with each passing quarter.
- From Q1 to Q4, the total number of orders decreased by 111.

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Question 7: Calculate the net revenue generated by the company. What is the quarter-over-quarter % change in net revenue?

Solution Query:

```
WITH QREV AS (
   SELECT
      quarter_number,
      ROUND(SUM(quantity * (vehicle_price - (discount * vehicle_price))), 2) AS REVENUE
   FROM order_t
   GROUP BY quarter_number
)

SELECT
   quarter_number,
   REVENUE,
   ROUND(LAG(REVENUE) OVER (ORDER BY quarter_number), 2) AS PREVIOUS_QUARTER,
   ROUND(
      (REVENUE - LAG(REVENUE) OVER (ORDER BY quarter_number))
      / LAG(REVENUE) OVER (ORDER BY quarter_number),
      2
   ) AS QOQ_PERCENTAGE
FROM QREV
ORDER BY quarter_number;
```

Output:

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```sql
WITH QREV AS (
    SELECT
        quarter_number,
        ROUND(SUM(quantity * (vehicle_price - (discount * vehicle_price))), 2) AS REVENUE
    FROM order_t
    GROUP BY quarter_number
)

SELECT
    quarter_number,
    REVENUE,
    ROUND(LAG(REVENUE) OVER (ORDER BY quarter_number), 2) AS PREVIOUS_QUARTER,
    ROUND(
        (REVENUE - LAG(REVENUE) OVER (ORDER BY quarter_number))
        / LAG(REVENUE) OVER (ORDER BY quarter_number),
        2
    ) AS QOQ_PERCENTAGE
FROM QREV
ORDER BY quarter_number;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

quarter_number	REVENUE	PREVIOUS_QUARTER	QOQ_PERCENTAGE
1	18032549.90	NULL	NULL
2	13122995.76	18032549.90	-0.27
3	8882298.84	13122995.76	-0.32
4	8573149.28	8882298.84	-0.03

Result 1

Output

Action Output

#	Time	Action	Message
1	21:50:03	use vehdb	0 row(s) affected
2	21:50:17	WITH QREV AS (SELECT quarter_number, ROUND(SUM(quantity * (vehicle_price - (discount * vehicle_price))), 2) AS RE...	4 row(s) returned

Observations and Insights:

- There's also a continuous decline in revenue when we compare the current and last quarter revenue.
- From Q1- Q2,it declined by 27%, Q2- Q3 it's 32% and in the last quarter it only dropped by 3% compared to Q3.

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Question 8: What is the trend of net revenue and orders by quarters?

Solution Query:

```
SELECT
   quarter_number,
   COUNT(order_id) AS Total_Orders,
   ROUND(
      SUM(
         quantity * (vehicle_price - (discount * vehicle_price))
      ),
      2
   ) AS REVENUE
FROM
   order_t
GROUP BY
   quarter_number
ORDER BY
   quarter_number;
```

Output:

```
SELECT
    quarter_number,
    COUNT(order_id) AS Total_Orders,
    ROUND(
        SUM(
            quantity * (vehicle_price - (discount * vehicle_price))
        ),
        2
    ) AS REVENUE
FROM
    order_t
GROUP BY
    quarter_number
ORDER BY
    quarter_number;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

quarter_number	Total_Orders	REVENUE
1	310	18032549.90
2	262	13122995.76
3	229	8882298.84
4	199	8573149.28

Result 1

Output

Action Output

#	Time	Action	Message
1	21:51:39	use vehdb	0 row(s) affected
2	21:52:00	SELECT quarter_number, COUNT(order_id) AS Total_Orders, ROUND(SUM(quantity * (vehicle_price - (discount * v...	4 row(s) returned

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Observations and Insights:

- Net revenue is also decreasing with the decreasing number of orders by each quarter.
- Total revenue of Q4 decreased by $9459401 compared to Q1.

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Question 9: What is the average discount offered for different types of credit cards?

Solution Query:

```
SELECT
   c.credit_card_type,
   ROUND(AVG(o.discount), 2) AS Avg_discount
FROM
   order_t o
INNER JOIN
   customer_t c
      ON o.customer_id = c.customer_id
GROUP BY
   c.credit_card_type
ORDER BY
   Avg_discount DESC;
```

Output:

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```sql
SELECT
    c.credit_card_type,
    ROUND(AVG(o.discount), 2) AS Avg_discount
FROM
    order_t o
INNER JOIN
    customer_t c
        ON o.customer_id = c.customer_id
GROUP BY
    c.credit_card_type
ORDER BY
    Avg_discount DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

credit_card_type	Avg_discount
laser	0.64
mastercard	0.63
visa-electron	0.62
china-unionpay	0.62
americanexpress	0.62
maestro	0.62

Result 1

Output

Action Output

#	Time	Action	Message
1	22:47:10	use vehdb	0 row(s) affected
2	22:47:10	SELECT c.credit_card_type, ROUND(AVG(o.discount), 2) AS Avg_discount FROM order_t o INNER JOI...	16 row(s) returned

Observations and Insights:

- Laser credit card has highest average discount(64%).
- Dinners-club-international has the lowest discount(58%)

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Question 10: What is the average time taken to ship the placed orders for each quarter?

Solution Query:

```
SELECT
   quarter_number,
   AVG(DATEDIFF(ship_date, order_date)) AS Avg_shipping_time
FROM
   order_t
GROUP BY
   quarter_number
ORDER BY
   quarter_number;
```

Output:

```
SELECT
    quarter_number,
    AVG(DATEDIFF(ship_date, order_date)) AS Avg_shipping_time
FROM
    order_t
GROUP BY
    quarter_number
ORDER BY
    quarter_number;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

quarter_number	Avg_shipping_time
1	57.1677
2	71.1107
3	117.7555
4	174.0955

Result 1

Output

Action Output

#	Time	Action	Message
1	21:56:06	use vehdb	0 row(s) affected
2	21:56:10	SELECT quarter_number, AVG(DATEDIFF(ship_date, order_date)) AS Avg_shipping_time FROM order_t GROUP B...	4 row(s) returned

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Business Metrics Overview

Total Revenue	Total Orders	Total Customers	Average Rating
$48610994	1000	994	3.1
Last Quarter Revenue	**Last quarter Orders**	**Average Days to Ship**	**% Good Feedback**
$8573149	199	98	21.5%

Business Recommendations

- Conduct Root Cause Analysis and Implement Service Upgrades by identifying the patterns of declining orders and rating.
- Prioritize high-volume states like Texas and California and focus on expansion over there. Like California market is evenly distributed, focus on Chevrolet promotion as it's already the most preferred vehicle maker by introducing competitive pricing. Also focus on states like Florida and New York as those states also have potentials to grow into a bigger market.
- Optimizing pricing and discount by partnering with credit card companies like Laser credit cards for exclusive deals and offers and higher discounts for repeat customers with positive reviews to encourage loyalty.
- Shipping timing needs to be improved by identifying the reason behind shipping delays.
- Track key metrics (ratings, feedback categories, orders, revenue) in a centralized dashboard. Set alerts for thresholds like ratings below 3.0 or revenue drops over 5%. Use this to adjust strategies dynamically.
- Conduct a survey of recent customers (especially those giving "Very bad" feedback) to understand issue points, such as vehicle quality, delivery delays, or post-sale support. Allocate resources to quality assurance for top brands like Chevrolet and Ford.

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